

Supplementary Information

A geometric model of democratic stability and the origin of the critical stress threshold $S_2 = \sqrt{3}/4$

This supplement provides (1) the complete geometric derivation, (2) the full bifurcation analysis, (3) the asymmetry conjecture with phenomenological illustration, (4) early-warning signal methodology, and (5) explicit discussion of all assumptions and their testability.

S1. Complete geometric derivation

S1.1 Vector placement

The three institutional pillars are placed as vectors in $\mathbb{R}^2$ separated by $120^\circ = 2\pi/3$ radians:

$$\begin{aligned} \mathbf{r}_R &= (R, 0) \\ \mathbf{r}_M &= (-M/2, (\sqrt{3}/2)M) \\ \mathbf{r}_N &= (-N/2, -(\sqrt{3}/2)N) \end{aligned} \quad (S1)$$

with $R, M, N \in [0, 1]$.

S1.2 Verification of 120° separation

The angle between $\mathbf{r}_R$ and $\mathbf{r}_M$ satisfies:

$$\cos \theta_{RM} = (\mathbf{r}_R \cdot \mathbf{r}_M) / (|\mathbf{r}_R| |\mathbf{r}_M|) = [R(-M/2)] / (RM) = -1/2$$

Hence $\theta_{RM} = \arccos(-1/2) = 120^\circ$. By symmetry, all pairwise angles equal 120° . $\square$

S1.3 Complete area derivation

Side vectors:

$$\begin{aligned} \mathbf{u} &= \mathbf{r}_M - \mathbf{r}_R = (-M/2 - R, (\sqrt{3}/2)M) \\ \mathbf{v} &= \mathbf{r}_N - \mathbf{r}_R = (-N/2 - R, -(\sqrt{3}/2)N) \end{aligned} \quad (S2)$$

The signed area is computed from $u_x v_y - u_y v_x$:

$$\begin{aligned} u_x v_y &= (-M/2 - R)(-(\sqrt{3}/2)N) = (\sqrt{3}/4)MN + (\sqrt{3}/2)RN \\ u_y v_x &= ((\sqrt{3}/2)M)(-N/2 - R) = -(\sqrt{3}/4)MN - (\sqrt{3}/2)MR \\ u_x v_y - u_y v_x &= (\sqrt{3}/2)MN + (\sqrt{3}/2)RN + (\sqrt{3}/2)MR \end{aligned}$$

Since $R, M, N \geq 0$, this expression is non-negative. The triangle area is half the absolute value:

$$A(R, M, N) = (1/2)(\sqrt{3}/2)(RM + RN + MN) = (\sqrt{3}/4)(RM + RN + MN) \quad (S3)$$

S1.4 Maximum area

With $R = M = N = 1$:

$$A_{\max} = A(1, 1, 1) = (\sqrt{3}/4)(1 + 1 + 1) = 3\sqrt{3}/4 \quad (S4)$$

This is also the area of an equilateral triangle with side length $\sqrt{3}$:

$$||r_R - r_M||_{\{R=M=1\}} = \sqrt{[(1 + 1/2)^2 + (0 - \sqrt{3}/2)^2]} = \sqrt{(9/4 + 3/4)} = \sqrt{3}$$

$$\text{Area} = (\sqrt{3}/4)(\sqrt{3})^2 = 3\sqrt{3}/4$$

S1.5 Normalised stability and stress

$$\Phi = A/A_{\max} = (RM + RN + MN)/3 \in [0,1] \quad (S5)$$

$$S = 1 - \Phi = 1 - (RM + RN + MN)/3 \in [0,1] \quad (S6)$$

S2. Derivation of S_1 and S_2

S2.1 $S_1 = \sqrt{3}/8$

The midpoint on the stress scale between full stability and the critical threshold is:

$$S_1 = S_2/2 = \sqrt{3}/8 \approx 0.2165 \quad (S7)$$

S_1 is a conventional early-warning level, not deductively derived.

S2.2 $S_2 = \sqrt{3}/4$

The area of a unit equilateral triangle (side length 1) is $\sqrt{3}/4$. In the normalised stress scale:

$$S_2 = \sqrt{3}/4 \approx 0.4330 \quad (S8)$$

At this stress, $\Phi_c = 1 - \sqrt{3}/4 \approx 0.567$ and $A_c = (3\sqrt{3}/4)(1 - \sqrt{3}/4) \approx 0.737$. S_2 follows deductively from the geometry of the three-pillar representation.

S3. Complete bifurcation analysis

S3.1 Potential function

$$U(\Phi;S) = (1/2)k(S)(\Phi - \Phi_c)^2 + (\lambda/4)(\Phi - \Phi_c)^4, \quad \lambda > 0 \quad (S9)$$

with $k(S) = \alpha(S_2 - S)$, $\alpha > 0$, and $\Phi_c = 1 - S_2$.

S3.2 Equilibrium and stability analysis

$$\partial U / \partial \Phi = (\Phi - \Phi_c)[k(S) + \lambda(\Phi - \Phi_c)^2] = 0 \quad (S10)$$

Solutions are $\Phi = \Phi_c$, which always exists; and, when $k(S) < 0$, $\Phi = \Phi_c \pm \sqrt{(-k(S)/\lambda)}$.

$$\partial^2 U / \partial \Phi^2 = k(S) + 3\lambda(\Phi - \Phi_c)^2 \quad (S11)$$

For $\Phi = \Phi_c$, $\partial^2 U / \partial \Phi^2 = k(S)$. Hence $S < S_2$ implies stability, $S = S_2$ implies marginal stability, and $S > S_2$ implies instability. For $\Phi = \Phi_c \pm \sqrt{(-k(S)/\lambda)}$, the second derivative is $-2k(S) > 0$, so both non-zero branches are stable local minima.

S3.3 Bifurcation type

The change in the number and stability of equilibria at $S = S_2$ constitutes a supercritical pitchfork bifurcation: for $S < S_2$ there is one stable equilibrium; at $S = S_2$ there is one marginally stable equilibrium; for $S > S_2$ there are two stable equilibria and one unstable equilibrium.

S3.4 Critical slowing down

Near S_2 , the dominant linear recovery rate below the threshold is proportional to $k(S)$. As S approaches S_2 from below, $k(S)$ approaches zero. This vanishing return rate is the mathematical signature of critical slowing down: perturbations decay increasingly slowly. The statistical consequences—rising variance and autocorrelation—are derived in Wissel (1984).

S4. The asymmetry conjecture: geometric motivation

S4.1 Empirical background

It is widely documented that democratic erosion can occur rapidly while recovery typically requires sustained, multi-dimensional reconstruction. This asymmetry—democracies are easier to break than to fix—motivates a quantitative conjecture within the geometric framework.

S4.2 Conjectural reasoning

The constant $\sqrt{3}$ appears in the geometric model in two capacities: as the side-length-related constant in the equilateral construction and as the structural constant associated with the threshold S_2 . The conjecture extends this constant to a third domain: the magnitude of recovery-erosion asymmetry.

The reasoning is analogical. If the post-bifurcation landscape—with its deeper lower branch (erosion) and shallower upper branch (recovery)—reflects the same structural imbalance that produces the $\sqrt{3}/4$ threshold, then the ratio of recovery effort to erosion magnitude might approximate $\sqrt{3}$.

This is not a deduction. Unlike S_2 , which follows from the geometric definition, $\mathcal{K} = \sqrt{3}$ is a conjectural extension. It could be false even if S_2 is correct.

S4.3 Phenomenological illustration

$$U_a(\Phi; S) = (1/2)k(S)(\Phi - \Phi_c)^2 + (\mu/3)(\Phi - \Phi_c)^3 + (\lambda/4)(\Phi - \Phi_c)^4 \quad (S12)$$

with $\mu > 0$ representing a bias favouring erosion. This potential produces unequal barrier heights for erosion and recovery transitions. Depending on the value of μ , the barrier ratio can take any value above unity. The imperfect pitchfork therefore demonstrates that the mathematical framework accommodates asymmetry, but it does not uniquely predict $\sqrt{3}$. The conjecture $\mathcal{K} = \sqrt{3}$ is not derived from this model; rather, the model illustrates the type of mechanism that could produce such asymmetry.

S4.4 Falsifiability

The conjecture is tested against five competing values: $\mathcal{K} = 1$ (symmetry null), $\sqrt{3}/2$, $\sqrt{3}$, $2\sqrt{3}$, and $3\sqrt{3}$. If $\sqrt{3}$ is the only value consistently within empirical confidence intervals across operationalisations, the conjecture survives falsification. If multiple values survive, the conjecture lacks specificity. If no value survives, the geometric approach to asymmetry is unsupported.

S4.5 Relationship to main text

The main text reports the empirical test in Section 4.6. The conjecture's status is distinct from S_2 : S_2 is deductively derived; $\mathcal{K}$ is empirically tested but not deductively derived. This distinction is maintained throughout.

S5. Early-warning signal methodology

S5.1 Rolling window computations

For window length w , the rolling mean at time t is:

$$\bar{S}_{it} = (1/w) \sum_{j=0}^{w-1} S_{i,t-j}$$

Detrending is performed via OLS within each window. Residuals are used for all subsequent computations.

S5.2 Variance

$$V_{it} = (1/(w-1)) \sum_{j=0}^{w-1} [\hat{\epsilon}_{i,t-j} - (1/w) \sum_{k=0}^{w-1} \hat{\epsilon}_{i,t-k}]^2$$

S5.3 Lag-1 autocorrelation

$$AR1_{it} = \frac{\sum_{j=1}^{w-1} (e_j^{(1)} - \bar{e}^{(1)})(e_j^{(2)} - \bar{e}^{(2)})}{\sqrt{[\sum_{j=1}^{w-1} (e_j^{(1)} - \bar{e}^{(1)})^2 \sum_{j=1}^{w-1} (e_j^{(2)} - \bar{e}^{(2)})^2]}}$$

where $e^{(1)}$ and $e^{(2)}$ are residual vectors lagged appropriately.

S6. Assumptions and their testability

Assumption	Content	How to test
A1: Equal fundamentality	R, M, N equally important	Weight pillars unequally; compare predictive performance
A2: Area representation	Capacity proportional to triangle area	Test alternative geometries: perimeter, product, sum
A3: Symmetric bifurcation	Pitchfork structure at S_2	Test alternative bifurcation types
A4: 120° separation	Pillars equally spaced	Test alternative angles; compare fit
A5: Phenomenological potential	Equation (7) describes dynamics	Estimate structural parameters; compare alternatives

All assumptions are explicit and, in principle, testable. This is what distinguishes the framework from numerology.

Supplementary References

1. Wissel, C. A universal law of the characteristic return time near thresholds. *Oecologia* 65, 101–107 (1984).
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3. Dakos, V. et al. Methods for detecting early warnings of critical transitions. *PLoS ONE* 7, e41010 (2012).
4. Strogatz, S. H. *Nonlinear Dynamics and Chaos*. Westview Press, 2nd ed. (2015).